

Chapter 1: Introduction to Technopreneurship

Entrepreneurs Are Made and Not Born

"The entrepreneurial mystique? It's not magic, it's not mysterious and it has nothing to do with the genes. It is a discipline. And like any discipline, it can be learned." – Peter Drucker

Who is an Entrepreneur?

An innovator or developer who recognizes and seizes opportunities; converts these opportunities into workable/marketable ideas; adds value through time, effort, money or skills; assumes the risks of the competitive marketplace to implement these ideas; realizes the rewards from these efforts. – (Kuratko & Hodgetts, 2007)

What is Entrepreneurship?

- Entrepreneurship is the practice of embarking on a new business or reviving an existing business by pooling together a bunch of resources in order to exploit new found opportunities.
- Creating the "New" and destroying the "Old":
 - New knowledge
 - New products
 - New processes
 - New services
 - New markets
 - New business models
 - New raw materials

Who is a Technopreneur?

- An entrepreneur who uses cutting-edge technology to develop new business. (Daniel Mankani, 2003).
- Is an entrepreneur who involves himself in technological changes in producing goods and services for his organization. (Sarimah Hanim & Abd. Rashid, 2008).
- They are entrepreneurs who used "technology" as their driven factor in transforming resources into goods and services, creating an environment conducive to industrial growth" ...

Types of Technology Entrepreneurship

2 Types of Technology Entrepreneurship

1. **Technology Developers** – those who develop unique technology capable of driving a new business (**INVENTORS**).
2. **Technology Users** – those who see a new technology and understand how it can be applied to meet a new market need (**INNOVATORS**).

Famous Technopreneurs

- Steve Jobs (Apple)
- Bill Gates (Microsoft)
- Mark Zuckerberg (Facebook)
- Sergey Brin and Larry Page (Google)

Technological Entrepreneurship

- It is simply entrepreneurship in a technology intensive context.
- It is a process of merging technology prowess and entrepreneurial talent and skills.

Examples of Technopreneurship

A technology-based enterprise is one that derives a competitive advantage from direct or indirect use of technology.

Example 1: Apple iPod

- it has a number of innovations
- the most significant of which is 160Gb 1.8-inch hard disk drive (smaller than average hard disk 3.5 inch a 40Gb capacity)
- Many people, even techies were fascinated by this and wonder how Apple able to do that.

Example 2: UPS (United Parcel Services)

- able to offer one-day delivery anywhere in the world.
- They have their sorting and delivery facility that could process up to 1,000,000,000 parcels a year
- It doesn't limit technopreneurship to really high-tech stuff or to business we normally relate to as 'technology-based'.
- For example, an internet café is not considered a technology-based enterprise even if it uses computers in its business operation.

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- However, it could be considered a technology-based enterprise if the computers and or the computer network is configured much differently than ordinary internet café businesses.

Who is an Intrapreneur?

- Are individuals in organizations with high entrepreneurial characteristics. – (Pinchot 1985).
- Is an entrepreneur within a large organization with high entrepreneurial characteristics who strongly believes in his talent and ability and has a strong desire to create something using his own initiative and creativity. – (Sarimah Hanim & Abd. Rashid, 2008).
- In the world corporate business, intrapreneurs emerge as that breed who is a cross between managers and entrepreneurs.
- They work for the corporation but are given the task of starting new ventures.

Sony PlayStation

Intraprenur: Ken Kutaragi

Company: Sony Computer Entertainment Inc.

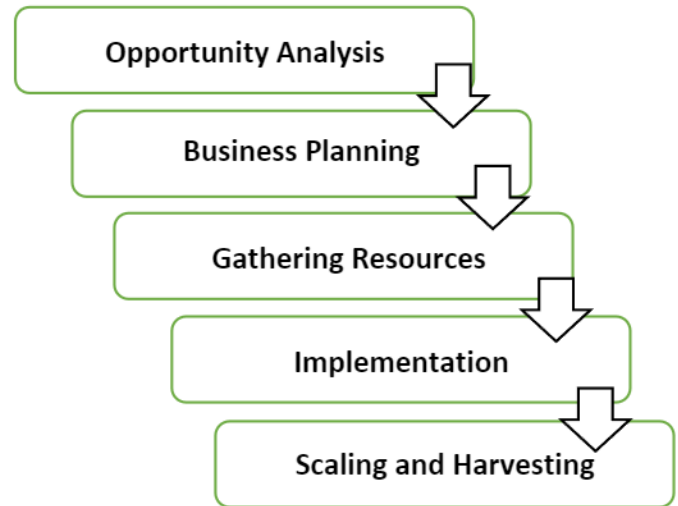
Year Launched: 1994

- Ken Kutaragi was working in Sony's sound labs when he bought his daughter a Nintendo game console. Watching her play, he was dismayed by the system's primitive sound effects.
- He realized that a digital chip dedicated solely to sound would improve the quality of the games and the product itself.
- Keeping his job at Sony, Kutaragi developed the SPC7000 for the next generation of Nintendo machines.
- Sony execs nearly fired him after discovering his sideline project, but then-CEO Norio Ohga realized the value of his innovation and encouraged Kutaragi's efforts.
- With Sony's blessing, Kutaragi worked with Nintendo to develop a CD-ROM-based Nintendo.
- Kutaragi helped Sony develop its own gaming system, which became the PlayStation.
- The first PlayStation made Sony a major player in the games market, but the

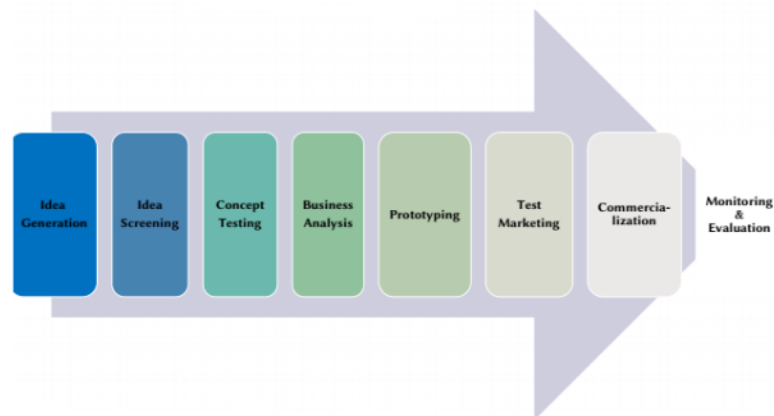
PlayStation 2 did even better, becoming the best-selling game console of all time.

Entrepreneurial Process vs. Technopreneurial Process

Entrepreneurial Process



Technopreneurial Process



- Idea Generation
- Idea Screening
- Concept Testing
- Business Analysis
- Prototyping
- Test Marketing
- Commercialization
- Monitoring & Evaluation

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Entrepreneurship vs. Technopreneurship

Similarities

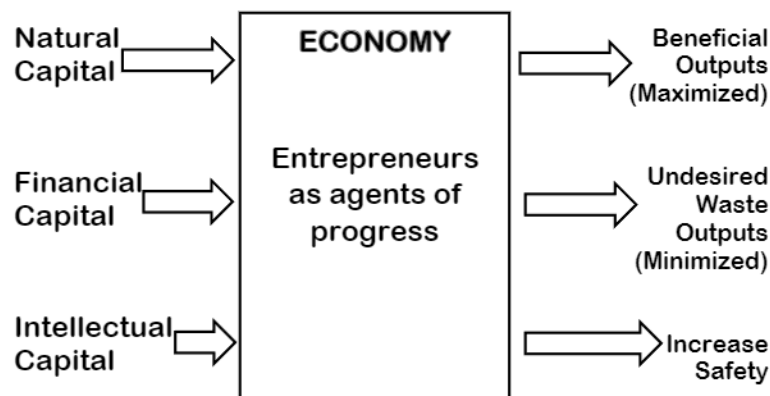
- Able to determine risk and has the courage to take risks
- Independent and self-confident, yet knows where to get help
- Likes a challenge
- Hardworking and willing to stick with a project
- Not easily discouraged
- Robust, very energetic and can handle stress
- Has a strong sense of self-worth.
- A positive thinker who does dwell on setbacks
- Often has a close friend or relative who owns a business

Differences

An Entrepreneur	A Technopreneur
Likes to compete	Likes to innovate
Is a self-starter/pioneer	Is part of a team
Is able to do many things at once	Is able to do many things at once, but chooses to delegate
Is creative, and has dreams and goals	Is innovative and has a greater vision
Likes to work for him or herself and be in control	Likes to be the one to control innovation and be part of an evolution
Is motivated by a strong desire to achieve and attain financial success	Is motivated by a strong vision and the passion to innovate
Focuses his/her attention on the chances of success rather than the possibility of failure	Takes failure in stride and knows it will lead to success if correction can be made

Technopreneur and the Economy

- Technopreneurs as agents of economic growth



- **Three Elements of the Intellectual Capital**
 - **Human Capital** – the skills, capabilities and knowledge of the firm's people.
 - **Organizational Capital** – the patents, technologies, processes, databases and networks.
 - **Social Capital** – the quality of the relationships with customers, suppliers and partners.

Successful Traits of a Technopreneur

- Integrity
- Leadership
- Impatient: bias toward action (with analysis)
- Quick Clock Speed
- Modest Ego – seeks and accepts coaching
- Willing to be different but knows it (not oblivious).
- Pragmatic: willing to compromise
- Rejoices in other's victories
- Driven to solve a valuable problem for customers
- Strong entrepreneurial intensity
- Willingness to incur the costs of growth
- Willingness to use wide range of financing sources
- Emphasis on a team-based organizing structure
- Focus on innovation
- Committed to commercialization of technology discovery
- Excellent communication skills

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- Understand the value of business principles
 - Formation and execution of a sound business plan
 - Raising of money
 - Building an organizational team

Factors Leading to Entrepreneurial Success

- Ability to initiate and operate a purposeful enterprise
- Ability to operate within the context and industrial environment at the time of initiation
- Ability to identify and screen timely opportunities
- Ability to accumulate and manage knowledge and technology
- Ability to mobilize resources – financial, physical and human
- Ability to assess and mitigate uncertainty and risk associated with the initiation of the enterprise
- Ability to provide and innovative contribution that encompasses novelty and originality
- Ability to encourage a collaborative team of people who have the capabilities and knowledge necessary for success

Technopreneurship Development in the Philippines

- Turning Up Talent: 'Technopreneurs' Drive Development in the Philippines
- Strengthening Higher Education – United States Agency for International Development's (USAID) *Innovative Development through Entrepreneurship Acceleration* (IDEA) project works with Philippine higher education engineering and science programs to inject entrepreneurship in their curriculum and allow students to transform their innovative ideas into products and businesses.
- Building Businesses – Global Entrepreneurship Symposium and Workshop

- Higher education is incorporating entrepreneurship into curriculums to supply graduates—and the economy with the skills needed to answer market demands.

Technopreneurs in the Philippines

Diosdado (Dado) Banatao

- was born to a rice farmer and housekeeper and grew up in Barrio Malabbac in the farming town of Iguig.
- Graduate cumlaude at Mapua Institute of Technology (electrical engineering degree).
- gave the world the first Ethernet controller chip
- the first chip set that significantly reduced the complexity of the personal computer
- He was honored with the prestigious Ernst and Young "Master Entrepreneur of the Year" award in the US.
- Forbes Magazine has often included him on its "Midas List"
- With \$5000,000 pooled from friends, he started Mostron which came up with the first system logic chipset for the XT and the AT personal computers, which lowered the cost of building the PC while making it a powerful machine.
- In 1996, his second startup "Chips and Technologies" was bought by Intel for a reported \$430 million.
- Founded Philippine Development Foundation (PhilDev), to promote science, technology, research and entrepreneurship that would advance long-term economic growth and eradicate poverty in the Philippines

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Winston Damarillo

- finished his industrial and mechanical engineering degree at De La Salle University
 - gained experience as an engineer before trying his luck in Silicon Valley
 - After a lot of attempts, he was able to make it as an employee in one of the top IT companies in the world.
 - At 30 years old he was able to establish his own company, which he sold to IBM for around \$100 million.
 - Today, he is an active member of PhilDev as its trustee and mentor.
 - Winston believes that what young Filipinos need are good mentors who will guide them as they find their place in the field of technology, just as he was mentored by his former colleagues when he was still starting out.
 - More importantly, he shares that success can be achieved if one is pursuing something he is truly passionate about.
- At that time, cellular phones were way too expensive, and this was an opportunity for Smart to transform it into a simple and affordable consumer item.
 - The venture became a success, and soon they sold the company to PLDT. Doy then involved himself in a number of companies that invested in IT start-ups, but went back to Smart as its chief wireless advisor.

Orlando B. Veal

- co-founder of Smart Communications.
- At one point in his life, he even got a scholarship in physics to be a scientist. But then he changed his mind and took up economics instead, from which he graduated cum laude.
- Doy went on to become a banker after graduation, but soon enough went back to the telecoms industry.
- Together with the late Dave Fernando, he founded Smart Communications, a company that simplified communication between Filipinos through the handheld phone.